

Curriculum Vitae

Research Interests

Cryogenic Energy Storage, Thermal Energy Conversion and Storage, Heat and Mass Transfer, Energy Systems Modelling, Digital Twins, Data-Driven Modelling, Engineering Design Optimization

Education

- 2021–Present **Ph.D. in Cryogenic Engineering**, *Indian Institute of Technology Kharagpur*
Dissertation topic: Packed Bed Cryogenic Energy Recovery Systems
Supervisor: Prof. Pavitra Sandilya
- 2019–2021 **Master of Technology in Mechanical Engineering**, *National Institute of Technology Rourkela*
Specialization : *Cryogenics and Vacuum Technology*
Thesis topic : *Transient numerical analysis of chilldown phenomena in an LN2 transfer line*
- 2014–2018 **Bachelor of Technology in Mechanical Engineering**, *Raghu Engineering College*
Affiliated to Jawaharlal Nehru Technological University Kakinada

Research Experience

- 2021–Present **Doctoral Researcher**, *Cryogenic Engineering Centre, IIT Kharagpur*
- Developed computational models to investigate transient heat transfer and phase change phenomena in packed-bed cryogenic energy storage systems.
 - Proposed optimization frameworks for the design of packed-bed cryogenic energy storage systems and systematic selection of phase change materials.
 - Developed machine learning models for state-of-charge prediction and digital twin development of thermal energy storage systems.
 - Designed, fabricated, and experimentally validated cryogenic packed-bed test facilities for model validation and performance characterization.

Journal Articles

- 1 **Chamarthi, B.S.**, Sandilya, P. A Multi-Criteria Decision Analysis to Select Phase Change Materials for Packed Bed Cryogenic Energy Recovery Systems.
Journal of Energy Storage, **143** (2026), 119780.
 - 2 **Chamarthi, B.S.**, Sandilya, P. Exergy analysis of cold recovery in a cryogenic packed bed system
Indian Journal of Cryogenics, **48(1)** (2023). 16-22
 - 3 **Chamarthi, B.S.**, Sandilya, P. A preliminary analysis to determine the charging cutoff temperature in a packed bed cryogenic energy storage system
Indian Journal of Cryogenics, **49(1)** (2026). 61-66
- Under Review Rajendra Kumar, **Chamarthi, B.S.**, Sandilya, P. Numerical Evaluation of Packed Bed Cryogenic Energy Recovery System Using a Two-Dimensional Local Thermal Equilibrium Model
Journal of Energy Storage

Conference Proceedings

- 1 **Chamarthi, B.S.**, Sandilya, P. Conceptual Genetic Algorithm-Based Optimization of the Layer Heights in a Cascaded Packed Bed for Cryogenic Energy Storage

Proceedings of the ASME International Mechanical Engineering Congress and Exposition (IMECE), Portland, OR, USA, 2024.

- 2 **Chamarthi, B.S.,** Sandilya, P. A Comparative Study on the Machine Learning Based Prediction of the State of Charge of Packed Beds for Cryogenic Energy Recovery
IOP Conference Series: Materials Science and Engineering, Advances in Cryogenic Engineering

Conference Presentations

- 1 **ASME International Mechanical Engineering Congress and Exposition 2024**, Portland, OR, USA – *Oral*
- 2 **21st International Conference on Sustainable Development in Energy, Water and Environmental Systems 2026**, Gran Canaria, Spain – *Oral*
“Second-Law Performance Evaluation of PCM-Based Packed Beds for Cryogenic Energy Recovery”
- 3 **28th National Symposium on Cryogenics and Superconductivity 2022**, IIT Kharagpur, Kharagpur, India –*Poster*
- 4 **International Conference on Sustainable Development in Chemical and Environmental Engineering 2024**, Thapar University, Punjab, India –*Poster*
“Selection of energy storage materials for cryogenic energy recovery”
- 5 **30th International Cryogenic Engineering Conference/ International Cryogenic Materials Conference 2026**, Daejeon, Republic of Korea – *Poster*

Technical Expertise

Computational Modelling	Transient and coupled heat and mass transfer, Phase Change Systems, Local Thermal Non Equilibrium (LTNE), Packed-Bed Thermal Energy Storage
Optimization	Multi-Criteria Decision Analysis (MCDA), Genetic Algorithms (GA), Design Optimization
Machine Learning	Time series prediction, Deep Learning, Regression Models
Experimental	Design and Fabrication of cryogenic experimental rigs, Temperature, Pressure and flow rate Instrumentation, Data Acquisition

Software Proficiency

Programming	MATLAB, Python
Simulation	ANSYS, ASPEN HYSYS
Design	SolidWorks
Fluid Property Libraries	REFPROP, CoolProp
Machine Learning Libraries	TensorFlow, Keras and Scikit-learn
Scientific Writing	LaTeX
Data analysis	OriginPro

Awards

- 2026 Travel Grant (1000 CHF) awarded by the ICEC30/ICMC2026 Organizing Committee
- 2024 Best Poster Award, SDCEE 2024.

- 2021-26 Institute research fellowship, MoE, Govt. of India, for Ph.D studies at IIT Kharagpur
2019-21 AICTE Postgraduate Scholarship (GATE-qualified) for M.Tech studies.

Teaching Experience

- 2021-25 **Teaching Assistant**, Cryogenic Engineering Centre, IIT Kharagpur
Courses Cryogenic Mass Transfer, Safety Issues with Cryogenic Fluids, LNG Engineering and Technology, Cryogenic Gas Separation
○ Assisted in preparing and evaluating assignments and examinations.
- 2021-23 **Teaching Assistant**, Swayam NPTEL, IIT Kharagpur
Courses Upstream LNG Technology, Mass, Momentum and Energy Balances in Engineering
○ Addressed course-related queries on the NPTEL discussion forum.
○ Assisted in preparing, grading assignments and examinations.